Ian Miller

miller42@engineering.upenn.edu | (440) 465-6007 | linkedin.com/in/ian-miller-8baa06368 | ianmiller093.github.io/

EDUCATION

University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA *May 2029*

Candidate for Bachelor of Science of Engineering *GPA*: 4.0/4.0

Major: Electrical Engineering

Relevant Coursework: ESE 2240: Signal Processing, ESE 2150: Circuits (Ongoing), ESE 2180: Devices (Ongoing), CIS 2400: Intro to Computer Systems (Ongoing), ESE 2030: Linear Algebra for Engineers, ESE 1120: Engineering E&M

Honors & Awards: UPenn Netter Center for Community Partnership Outstanding Volunteer Award

TECHNICAL SKILLS

Platforms/Tools: LTspice, KiCad, Arduino IDE, ESP32, Raspberry Pi, soldering, Java, Python, C++

Interests: Circuit design, PCB design, experimental physics/astronomy, power electronics, aerospace instrumentation

EXPERIENCE

Madhavacheril Lab Group, UPenn Department of Physics and Astronomy, Philadelphia, PA

Cosmology Researcher *May – August 2026*

- Built simulations of cosmic microwave background maps and used Gibbs Sampling methods for component separation.
- Sharpened signal processing and Bayesian inference intuition, developed communication and quick learning skills.
- Primary contributor to https://github.com/IanMiller093/cmb.
- Presented work at Haverford Summer Undergraduate Research Workshop and UPenn PURM Fall Research Expo.

Penn Reading Initiative, UPenn Netter Center for Community Partnership, Philadelphia, PA

Student Coordinator, Grades K-4 Reading at B. B. Comegys Elementary (Work Study) *August 2025 – Present*

- Hold one-on-one lessons 7-10 hours per week, individually catered lessons to children struggling with reading.
- Promoted to Head Tutor in second semester with the club, promoted to Student Coordinator in third semester.

PROJECTS

Locked Out: Student ID card attachment to never forget ID [ESE 1110 Final Project] *November – December 2025*

- Used MicroPython to program two Raspberry Pi Pico Ws to communicate location to each other using BLE.
- Configured sensor suite on door, including LED and buzzer to notify user along with servo to push deadbolt.

External Compensator 5 V -> 3.3 V Buck Converter PCB [Personal Project] [Ongoing] *August 2026*

- Followed TI buck converter IC datasheet to choose unique component values for my specific voltage stepdown goal.
- Tested component values in LTSpice, designed PCB using KiCad, will ship PCBs, solder and test real values.
- More info with writeup here: ianmiller093.github.io/

LEADERSHIP & ACTIVITIES

Penn Aerial Robotics (PennAiR), Philadelphia, PA

Electrical Team, In-Flight Sensors Subteam *August 2025 – Present*

- Wrote and debugged C++ code to control in flight LoRa communication of sensor values with control on ground.
- Worked in team of ~4 to integrate IMU, Hall-effect sensor, airspeed sensor, GPS, and others to sensor suite.

Access Engineering , Philadelphia, PA

Head TA of Electrical Engineering Team *August 2025 – Present*

- Became Head TA of the electrical engineering team in my second month in the club.
- Co-designed series of lessons for groups of ~25, total ~100 local Philly high schoolers per semester. Led team of ~10 TAs.